(5) In the case of the edge stiffeners of lipped C-sections and lipped Z-sections, C_θ should be determined with the unit loads u applied as shown in figure 5.6(c). This results in the following expression for the spring stiffness K_1 for the flange 1:

$$K_1 = \frac{E t^3}{4(1-\nu^2)} \cdot \frac{1}{b_1^2 h_w + b_1^3 + 0{,}5 b_1 b_2 h_w k_f} \qquad \text{… (5.10b)}$$

where:

b_1 is the distance from the web-to-flange junction to the gravity center of the effective area of the edge stiffener (including effective part b_{e2} of the flange) of flange 1, see figure 5.6(a);

b_2 is the distance from the web-to-flange junction to the gravity center of the effective area of the edge stiffener (including effective part of the flange) of flange 2;

h_w is the web depth;

$k_f = 0$ if flange 2 is in tension (e.g. for beam in bending about the y-y axis);

$k_f = \frac{A_{s2}}{A_{s1}}$ if flange 2 is also in compression (e.g. for a beam in axial compression);

$k_f = 1$ for a symmetric section in compression.

A_{s1} and A_{s2} is the effective area of the edge stiffener (including effective part b_{e2} of the flange, see figure 5.6(b)) of flange 1 and flange 2 respectively.

(6) For an intermediate stiffener, as a conservative alternative the values of the rotational spring stiffnesses $C_{\theta,1}$ and $C_{\theta,2}$ may be taken as equal to zero, and the deflection δ may be obtained from:

$$\delta = \frac{u b_1^{\,2} b_2^{\,2}}{3(b_1 + b_2)} \cdot \frac{12(1-\nu^2)}{E t^3} \qquad \text{... (5.11)}$$

(7) The reduction factor χ_d for the distortional buckling resistance (flexural buckling of a stiffener) should be obtained from the relative slenderness $\bar{\lambda}_d$ from:

$$\chi_d = 1{,}0 \qquad \text{if } \bar{\lambda}_d \le 0{,}65 \qquad \text{... (5.12a)}$$

$$\chi_d = 1{,}47 - 0{,}723\bar{\lambda}_d \qquad \text{if } 0{,}65 < \bar{\lambda}_d < 1{,}38 \qquad \text{... (5.12b)}$$

$$\chi_d = \frac{0{,}66}{\bar{\lambda}_d} \qquad \text{if } \bar{\lambda}_d \ge 1{,}38 \qquad \text{... (5.12c)}$$

where:

$$\bar{\lambda}_d = \sqrt{f_{yb} / \sigma_{cr,s}} \qquad \text{… (5.12d)}$$

where:

$\sigma_{cr,s}$ is the elastic critical stress for the stiffener(s) from 5.5.3.2, 5.5.3.3 or 5.5.3.4.

(8) Alternatively, the elastic critical buckling stress $\sigma_{cr,s}$ may be obtained from elastic first order buckling analysis using numerical methods (see 5.5.1(7)).

(9) In the case of a plane element with an edge and intermediate stiffener(s) in the absence of a more accurate method the effect of the intermediate stiffener(s) may be neglected.

5.5.3.2 Plane elements with edge stiffeners

(1) The following procedure is applicable to an edge stiffener if the requirements in 5.2 are met and the angle between the stiffener and the plane element is between 45° and 135°.

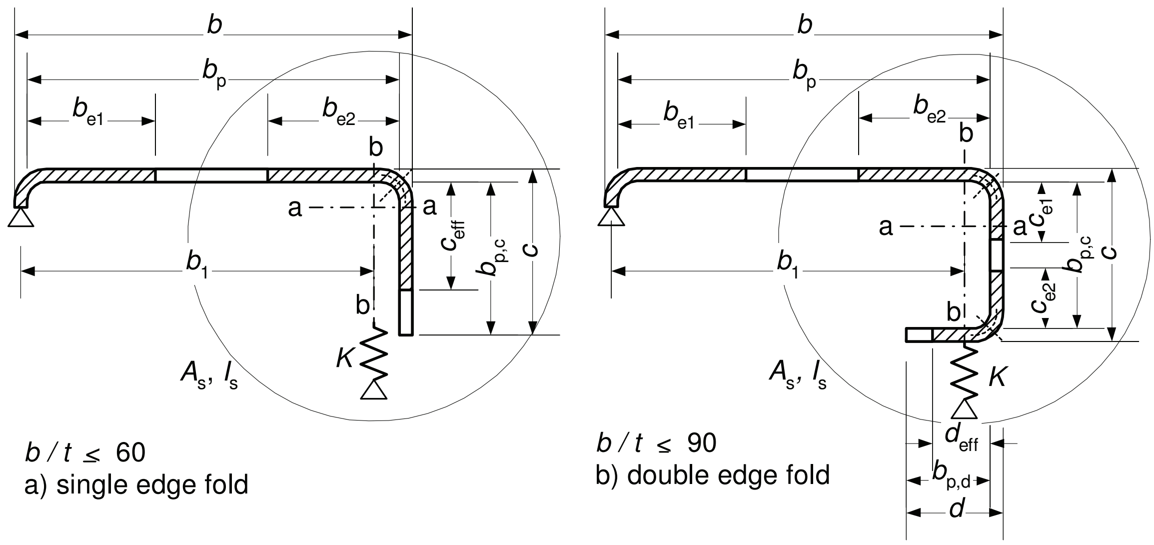


Figure 5.7: Edge stiffeners

(2) The cross-section of an edge stiffener should be taken as comprising the effective portions of the stiffener, element c or elements c and d as shown in figure 5.7, plus the adjacent effective portion of the plane element $b_{p.}$

(3) The procedure, which is illustrated in figure 5.8, should be carried out in steps as follows:

- **Step 1**: Obtain an initial effective cross-section for the stiffener using effective widths determined by assuming that the stiffener gives full restraint and that $\sigma_{com,Ed} = f_{yb}/\gamma_{M0}$, see (4) to (5);
- **Step 2**: Use the initial effective cross-section of the stiffener to determine the reduction factor for distortional buckling (flexural buckling of a stiffener), allowing for the effects of the continuous spring restraint, see (6), (7) and (8);
- **Step 3**: Optionally iterate to refine the value of the reduction factor for buckling of the stiffener, see (9) and (10).

(4) Initial values of the effective widths b_{e1} and b_{e2} shown in figure 5.7 should be determined from clause 5.5.2 by assuming that the plane element b_p is doubly supported, see table 4.1 in EN 1993-1-5.

(5) Initial values of the effective widths c_{eff} and d_{eff} shown in figure 5.7 should be obtained as follows:

a) for a single edge fold stiffener:

$$c_{eff} = \rho b_{p,c} \quad \text{... (5.13a)}$$

with ρ obtained from 5.5.2, except using a value of the buckling factor k_σ given by the following:

- if $b_{p,c}/b_p \leq 0{,}35$:

$$k_\sigma = 0{,}5 \quad \text{... (5.13b)}$$

- if $0{,}35 < b_{p,c}/b_p \leq 0{,}6$:

$$k_\sigma = 0{,}5 + 0{,}83 \sqrt[3]{(b_{p,c} / b_p - 0{,}35)^2} \quad \text{...(5.13c)}$$

b) for a double edge fold stiffener:

$$c_{eff} = \rho b_{p,c} \quad \text{... (5.13d)}$$

with ρ obtained from 5.5.2 with a buckling factor k_σ for a doubly supported element from table 4.1 in EN 1993-1-5

$$d_{eff} = \rho b_{p,d} \quad \text{... (5.13e)}$$

with ρ obtained from 5.5.2 with a buckling factor k_σ for an outstand element from table 4.2 in EN 1993-1-5.

(6) The effective cross-sectional area of the edge stiffener A_s should be obtained from:

$$A_s = t\,(b_{e2} + c_{eff}) \quad \text{or} \quad \text{... (5.14a)}$$

$$A_s = t\,(b_{e2} + c_{e1} + c_{e2} + d_{eff}) \quad \text{...(5.14b)}$$

respectively.

NOTE: The rounded corners should be taken into account if needed, see 5.1.

(7) The elastic critical buckling stress $\sigma_{cr,s}$ for an edge stiffener should be obtained from:

$$\sigma_{cr,s} = \frac{2\sqrt{K E I_s}}{A_s} \quad \text{... (5.15)}$$

where:

K is the spring stiffness per unit length, see 5.5.3.1(2).

I_s is the effective second moment of area of the stiffener, taken as that of its effective area A_s about the centroidal axis a - a of its effective cross-section, see figure 5.7.

(8) Alternatively, the elastic critical buckling stress $\sigma_{cr,s}$ may be obtained from elastic first order buckling analyses using numerical methods, see 5.5.1(7).

(9) The reduction factor χ_d for the distortional buckling (flexural buckling of a stiffener) resistance of an edge stiffener should be obtained from the value of $\sigma_{cr,s}$ using the method given in 5.5.3.1(7).

5.5.3.4 Trapezoidal sheeting profiles with intermediate stiffeners

5.5.3.4.1 General

(1) This sub-clause 5.5.3.4 should be used for trapezoidal profiled sheets, in association with 5.5.3.3 for flanges with intermediate flange stiffeners and 5.5.3.3 for webs with intermediate stiffeners.

(2) Interaction between the buckling of intermediate flange stiffeners and intermediate web stiffeners should also be taken into account using the method given in 5.5.3.4.4.

5.5.3.4.2 Flanges with intermediate stiffeners

(1) If it is subject to uniform compression, the effective cross-section of a flange with intermediate stiffeners should be assumed to consist of the reduced effective areas $A_{s,red}$ including two strips of width $0{,}5b_{eff}$ (or 15 t, see figure 5.11) adjacent to the stiffener.

(2) For one central flange stiffener, the elastic critical buckling stress $\sigma_{cr,s}$ should be obtained from:

$$\sigma_{cr,s} = \frac{4{,}2\,k_w E}{A_s}\sqrt{\frac{I_s\, t^3}{4\, b_p^2\,(2\, b_p + 3\, b_s)}} \qquad \text{... (5.22)}$$

where:

- b_p is the notional flat width of plane element shown in figure 5.11;
- b_s is the stiffener width, measured around the perimeter of the stiffener, see figure 5.11;
- A_s, I_s are the cross-section area and the second moment of area of the stiffener cross-section according to figure 5.11;
- k_w is a coefficient that allows for partial rotational restraint of the stiffened flange by the webs or other adjacent elements, see (5) and (6). For the calculation of the effective cross-section in axial compression the value $k_w = 1{,}0$.

The equation 5.22 may be used for wide grooves provided that flat part of the stiffener is reduced due to local buckling and b_p in the equation 5.22 is replaced by the larger of b_p and $0{,}25(3b_p+b_r)$, see figure 5.11. Similar method is valid for flange with two or more wide grooves.

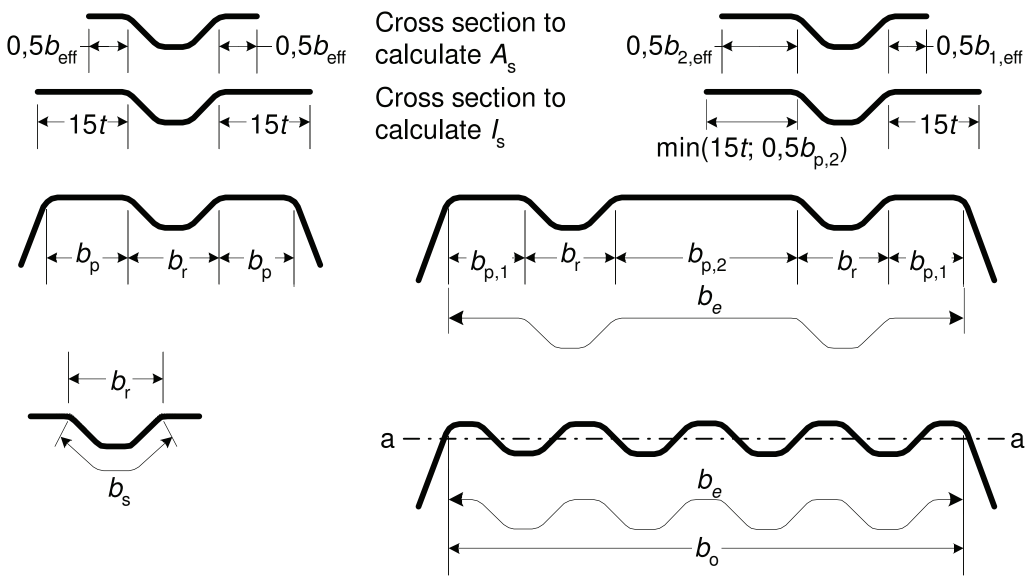


Figure 5.11: Compression flange with one, two or multiple stiffeners

(3) For two symmetrically placed flange stiffeners, the elastic critical buckling stress $\sigma_{cr,s}$ should be obtained from:

$$\sigma_{cr,s} = \frac{4{,}2\ k_w\ E}{A_s}\sqrt{\frac{I_s\ t^3}{8\ b_1^2\ (3\ b_e - 4\ b_1)}} \quad \text{... (5.23a)}$$

with:

$$b_e = 2b_{p,1} + b_{p,2} + 2b_s$$

$$b_1 = b_{p,1} + 0{,}5\,b_r$$

where:

$b_{p,1}$	is the notional flat width of an outer plane element, as shown in figure 5.11;
$b_{p,2}$	is the notional flat width of the central plane element, as shown in figure 5.11;
b_r	is the overall width of a stiffener, see figure 5.11;
A_s, I_s	are the cross-section area and the second moment of area of the stiffener cross-section according to figure 5.11.

(4) For a multiple stiffened flange (three or more equal stiffeners) the effective area of the *entire flange* is

$$A_{eff} = \rho b_e t \quad \text{…(5.23b)}$$

where ρ is the reduction factor according to EN 1993-1-5, Annex E for the slenderness $\overline{\lambda}_p$ based on the elastic buckling stress

$$\sigma_{cr,s} = 1{,}8E\sqrt{\frac{I_s\,t}{b_o^2 b_e^3}} + 3{,}6\frac{Et^2}{b_o^2} \quad \text{…(5.23c)}$$

where:

I_s	is the sum of the second moment of area of the stiffeners about the centroidal axis a-a, neglecting the thickness terms $bt^3/12$;
b_o	is the width of the flange as shown in figure 5.11;
b_e	is the developed width of the flange as shown in figure 5.11.

(5) The value of k_w may be calculated from the compression flange buckling wavelength l_b as follows:

- if $l_b / s_w \geq 2$:

$$k_w = k_{wo} \quad \text{...(5.24a)}$$

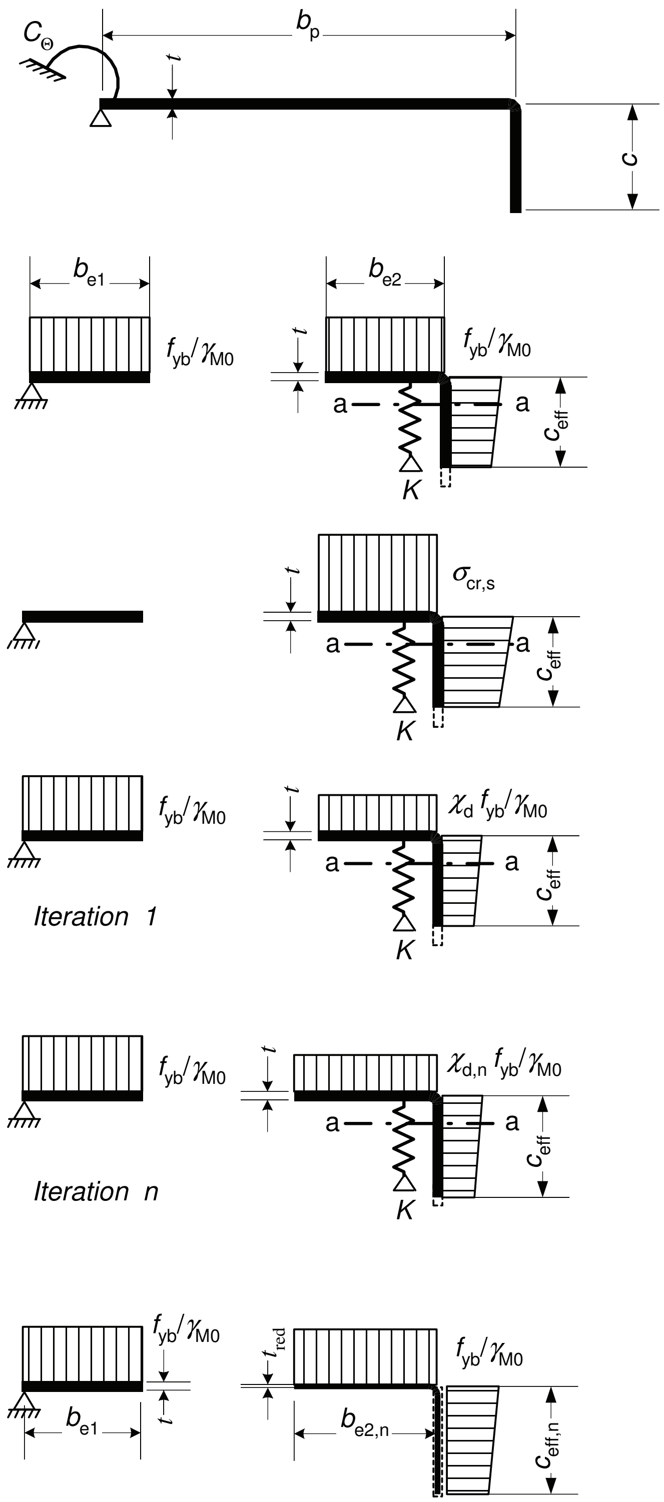
- if $l_b / s_w < 2$:

$$k_w = k_{wo} - (k_{wo} - 1)\left[\frac{2l_b}{s_w} - \left(\frac{l_b}{s_w}\right)^2\right] \quad \text{...(5.24b)}$$

where:

s_w	is the slant height of the web, see figure 5.1(c).

(6) Alternatively, the rotational restraint coefficient k_w may conservatively be taken as equal to 1,0 corresponding to a pin-jointed condition.



a) Gross cross-section and boundary conditions

b) **Step 1**: Effective cross-section for $K = \infty$ based on $\sigma_{com,Ed} = f_{yb}/\gamma_{M0}$

c) **Step 2**: Elastic critical stress $\sigma_{cr,s}$ for effective area of stiffener A_s from step 1

d) Reduced strength $\chi_d f_{yb}/\gamma_{M0}$ for effective area of stiffener A_s, with reduction factor χ_d based on $\sigma_{cr,s}$

e) **Step 3**: Optionally repeat step 1 by calculating the effective width with a reduced compressive stress $\sigma_{com,Ed,i} = \chi_d f_{yb} / \gamma_{M0}$ with χ_d from previous iteration, continuing until $\chi_{d,n} \approx \chi_{d,(n-1)}$ but $\chi_{d,n} \leq \chi_{d,(n-1)}$.

f) Adopt an effective cross-section with b_{e2}, c_{eff} and reduced thickness t_{red} corresponding to $\chi_{d,n}$

Figure 5.8: Compression resistance of a flange with an edge stiffener

(10) If $\chi_d < 1$ it may be refined iteratively, starting the iteration with modified values of ρ obtained using 5.5.2(5) with $\sigma_{com,Ed,i}$ equal to $\chi_d f_{yb}/\gamma_{M0}$, so that:

$$\overline{\lambda}_{p,red} = \overline{\lambda}_p \sqrt{\chi_d} \qquad \text{... (5.16)}$$

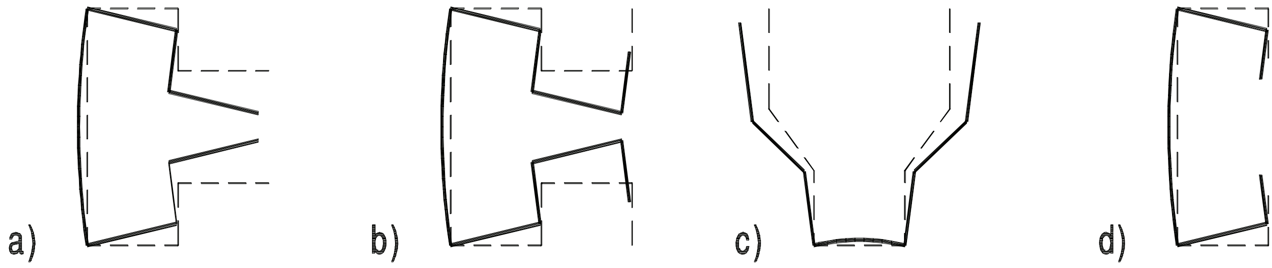


Figure 5.4: Examples of distortional buckling modes

(6) The effects of distortional buckling should be allowed for in cases such as those indicated in figures 5.4(a), (b) and (c). In these cases the effects of distortional buckling should be determined performing linear (see 5.5.1(7)) or non-linear buckling analysis (see EN 1993-1-5) using numerical methods or column stub tests.

(7) Unless the simplified procedure in 5.5.3 is used and where the elastic buckling stress is obtained from linear buckling analysis the following procedure may be applied:

1) For the wavelength up to the nominal member length, calculate the elastic buckling stress and identify the corresponding buckling modes, see figure 5.5a.
2) Calculate the effective width(s) according to 5.5.2 for locally buckled cross-section parts based on the minimum local buckling stress, see figure 5.5b.
3) Calculate the reduced thickness (see 5.5.3.1(7)) of edge and intermediate stiffeners or other cross-section parts undergoing distortional buckling based on the minimum distortional buckling stress, see figure 5.5b.
4) Calculate overall buckling resistance according to 6.2 (flexural, torsional or lateral-torsional buckling depending on buckling mode) for nominal member length and based on the effective cross-section from 2) and 3).

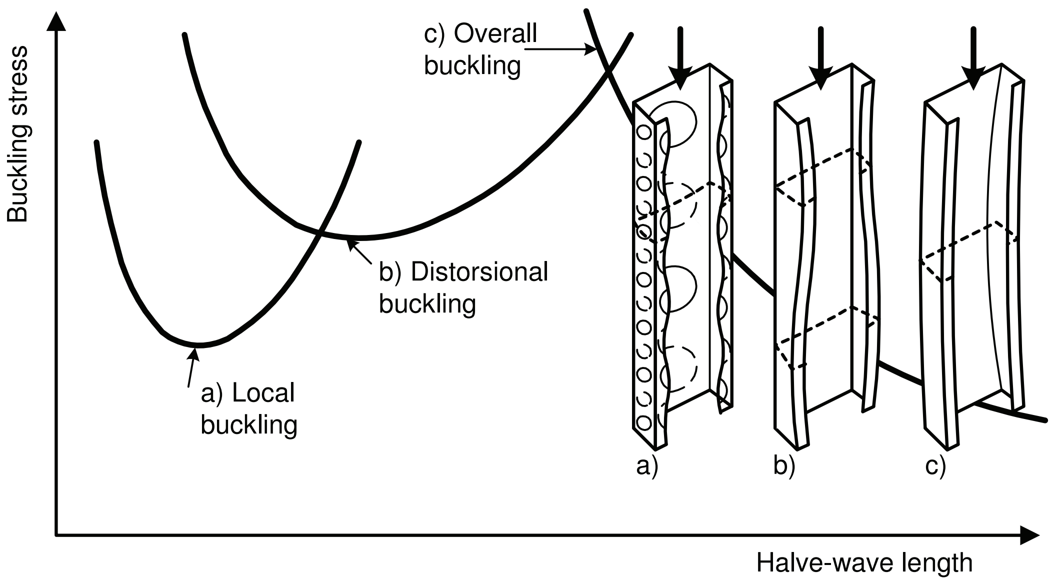


Figure 5.5a: Examples of elastic critical stress for various buckling modes as function of *halve-wave length* and examples of buckling modes.

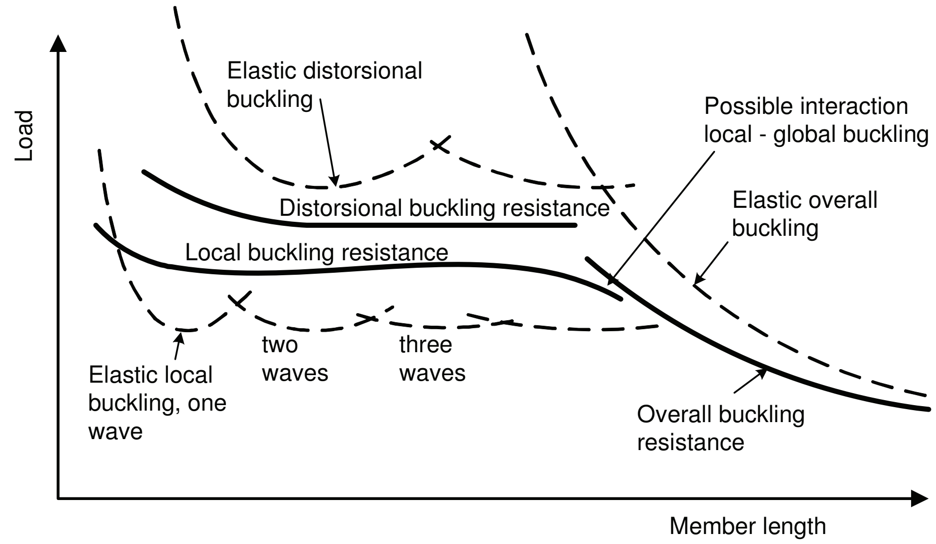


Figure 5.5 b: Examples of elastic buckling load and buckling resistance as a function of member length

5.5.2 Plane elements without stiffeners

(1) The effective widths of unstiffened elements should be obtained from EN 1993-1-5 using the notional flat width b_p for $\overline{b}$ by determining the reduction factors for plate buckling based on the plate slenderness $\overline{\lambda}_p$.

(2) The notional flat width b_p of a plane element should be determined as specified in figure 5.1 of section 5.1.4. In the case of plane elements in a sloping webs, the appropriate slant height should be used.

NOTE: For outstands an alternative method for calculating effective widths is given in Annex D.

(3) In applying the method in EN 1993-1-5 the following procedure may be used:

- The stress ratio ψ, from tables 4.1 and 4.2 used to determine the effective width of flanges of a section subject to stress gradient, may be based on gross section properties.
- The stress ratio ψ, from table 4.1 and 4.2 used to determine the effective width of web, may be obtained using the effective area of compression flange and the gross area of the web.
- The effective section properties may be refined by using the stress ratio ψ based on the effective cross-section already found in place of the gross cross-section. The minimum steps in the iteration dealing with the stress gradient are two.
- The simplified method given in 5.5.3.4 may be used in the case of webs of trapezoidal sheeting under stress gradient.

5.5.3 Plane elements with edge or intermediate stiffeners

5.5.3.1 General

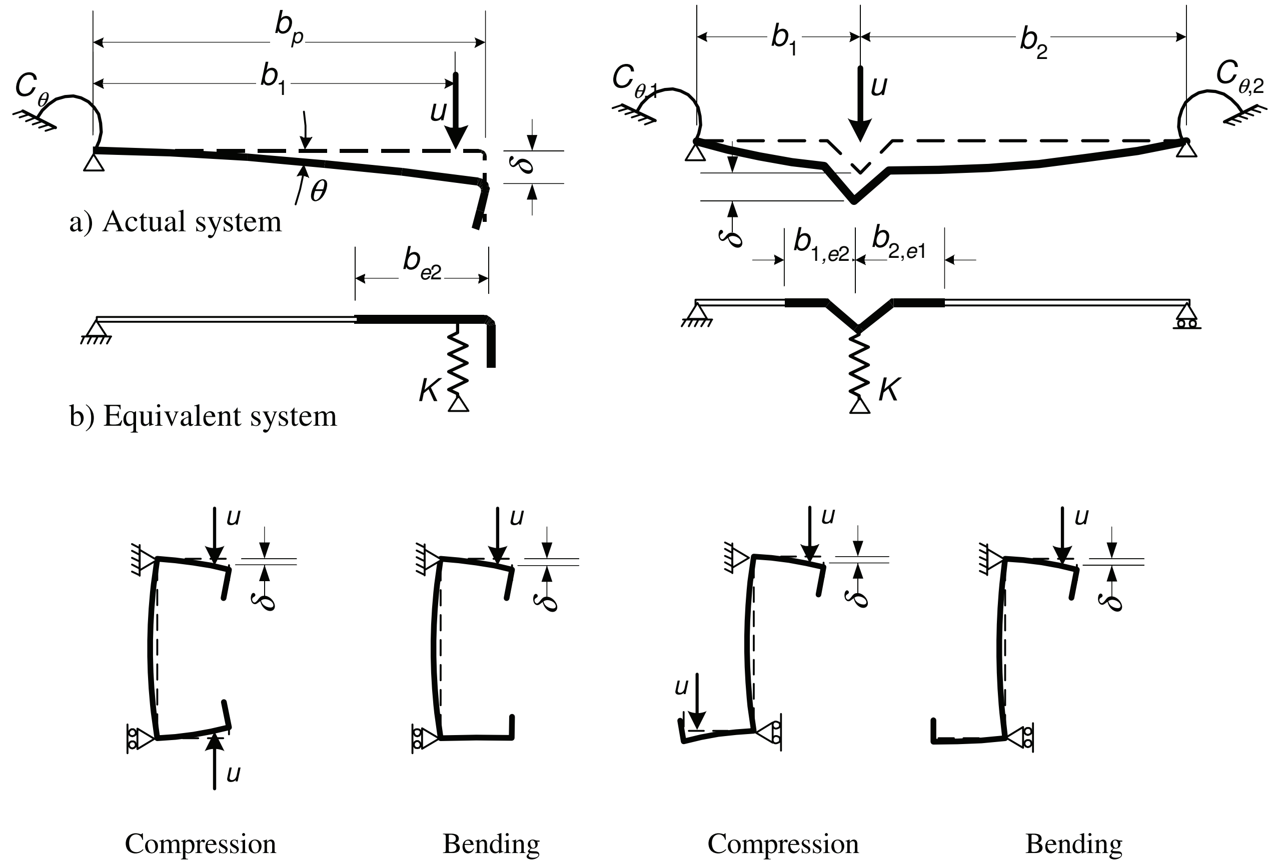
(1) The design of compression elements with edge or intermediate stiffeners should be based on the assumption that the stiffener behaves as a compression member with continuous partial restraint, with a spring stiffness that depends on the boundary conditions and the flexural stiffness of the adjacent plane elements.

(2) The spring stiffness of a stiffener should be determined by applying an unit load per unit length u as illustrated in figure 5.6. The spring stiffness K per unit length may be determined from:

$$K = u/\delta \qquad \text{... (5.9)}$$

where:

δ is the deflection of the stiffener due to the unit load u acting in the centroid (b_1) of the effective part of the cross-section.



c) Calculation of δ for C and Z sections

Figure 5.6: Determination of spring stiffness

(3) In determining the values of the rotational spring stiffnesses C_θ, $C_{\theta,1}$ and $C_{\theta,2}$ from the geometry of the cross-section, account should be taken of the possible effects of other stiffeners that exist on the same element, or on any other element of the cross-section that is subject to compression.

(4) For an edge stiffener, the deflection δ should be obtained from:

$$\delta = \theta\, b_{\mathrm{p}} + \frac{u b_{\mathrm{p}}^{\,3}}{3} \cdot \frac{12\left(1 - \nu^2\right)}{E t^3} \qquad \text{... (5.10a)}$$

with:

$$\theta = u\, b_{\mathrm{p}} / C_\theta$$

(11) The reduced effective area of the stiffener $A_{s,red}$ allowing for flexural buckling should be taken as:

$$A_{s,red} = \chi_d A_s \frac{f_{yb}/\gamma_{M0}}{\sigma_{com,Ed}} \qquad \text{but } A_{s,red} \le A_s \qquad \text{... (5.17)}$$

where

$\sigma_{com,Ed}$ is compressive stress at the centreline of the stiffener calculated on the basis of the effective cross-section.

(12) I n determining effective section properties, the reduced effective area $A_{s,red}$ should be represented by using a reduced thickness $t_{red} = t\,A_{s,red}/A_s$ for all the elements included in A_s.

5.5.3.3 Plane elements with intermediate stiffeners

(1) The following procedure is applicable to one or two equal intermediate stiffeners formed by grooves or bends provided that all plane elements are calculated according to 5.5.2.

(2) The cross-section of an intermediate stiffener should be taken as comprising the stiffener itself plus the adjacent effective portions of the adjacent plane elements $b_{p,1}$ and $b_{p,2}$ shown in figure 5.9.

(3) The procedure, which is illustrated in figure 5.10, should be carried out in steps as follows:

- **Step 1**: Obtain an initial effective cross-section for the stiffener using effective widths determined by assuming that the stiffener gives full restraint and that $\sigma_{com,Ed} = f_{yb}/\gamma_{M0}$, see (4) and (5);
- **Step 2**: Use the initial effective cross-section of the stiffener to determine the reduction factor for distortional buckling (flexural buckling of an intermediate stiffener), allowing for the effects of the continuous spring restraint, see (6), (7) and (8);
- **Step 3**: Optionally iterate to refine the value of the reduction factor for buckling of the stiffener, see (9) and (10).

(4) Initial values of the effective widths $b_{1,e2}$ and $b_{2,e1}$ shown in figure 5.9 should be determined from 5.5.2 by assuming that the plane elements $b_{p,1}$ and $b_{p,2}$ are doubly supported, see table 4.1 in EN 1993-1-5.

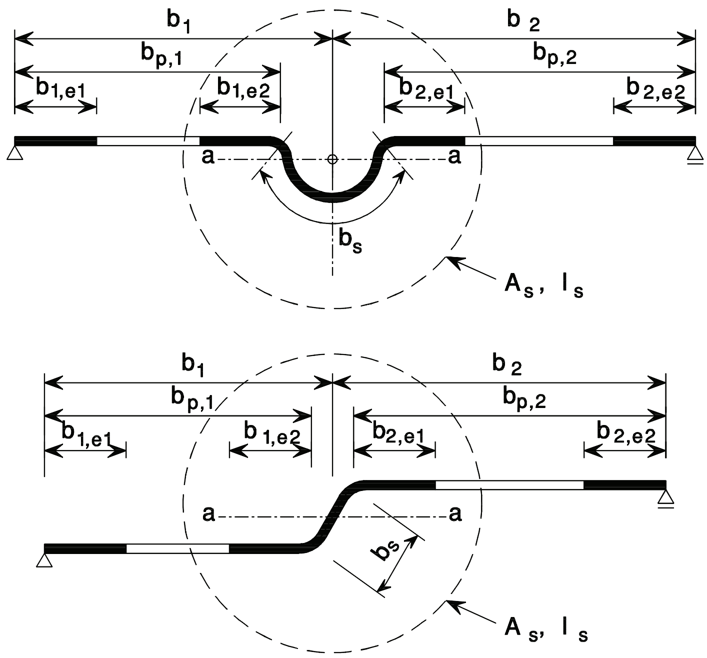


Figure 5.9: Intermediate stiffeners

(5) The effective cross-sectional area of an intermediate stiffener A_s should be obtained from:

$$A_s = t\,(b_{1,e2} + b_{2,e1} + b_s) \qquad \text{... (5.18)}$$

in which the stiffener width b_s is as shown in figure 5.9.

NOTE: The rounded corners should be taken into account if needed, see 5.1.

(6) The critical buckling stress $\sigma_{cr,s}$ for an intermediate stiffener should be obtained from:

$$\sigma_{cr,s} = \frac{2\sqrt{KEI_s}}{A_s} \qquad \text{... (5.19)}$$

where:

K is the spring stiffness per unit length, see 5.5.3.1(2).

I_s is the effective second moment of area of the stiffener, taken as that of its effective area A_s about the centroidal axis a - a of its effective cross-section, see figure 5.9.

(7) Alternatively, the elastic critical buckling stress $\sigma_{cr,s}$ may be obtained from elastic first order buckling analyses using numerical methods, see 5.5.1(7).

(8) The reduction factor χ_d for the distortional buckling resistance (flexural buckling of an intermediate stiffener) should be obtained from the value of $\sigma_{cr,s}$ using the method given in 5.5.3.1(7).

(9) If $\chi_d < 1$ it may optionally be refined iteratively, starting the iteration with modified values of ρ obtained using 5.5.2(5) with $\sigma_{com,Ed,i}$ equal to $\chi_d f_{yb} / \gamma_{M0}$, so that:

$$\bar{\lambda}_{p,red} = \bar{\lambda}_p \sqrt{\chi_d} \qquad \text{... (5.20)}$$

(10) The reduced effective area of the stiffener $A_{s,red}$ allowing for distortional buckling (flexural buckling of a stiffener) should be taken as:

$$A_{s,red} = \chi_d A_s \frac{f_{yb}/\gamma_{M0}}{\sigma_{com,Ed}} \quad \text{but } A_{s,red} \leq A_s \qquad \text{... (5.21)}$$

where

$\sigma_{com,Ed}$ is compressive stress at the centreline of the stiffener calculated on the basis of the effective cross-section.

(11) In determining effective section properties, the reduced effective area $A_{s,red}$ should be represented by using a reduced thickness $t_{red} = t\, A_{s,red} / A_s$ for all the elements included in A_s.

$$\delta = 0,43 \frac{\sum_{j=1}^{n} r_j \frac{\phi_j}{90°}}{\sum_{i=1}^{m} b_{p,i}} \qquad \ldots (5.1d)$$

where:

A_g	is	the area of the gross cross-section;
$A_{g,sh}$	is	the value of A_g for a cross-section with sharp corners;
$b_{p,i}$	is	the notional flat width of plane element i for a cross-section with sharp corners;
I_g	is	the second moment of area of the gross cross-section;
$I_{g,sh}$	is	the value of I_g for a cross-section with sharp corners;
I_w	is	the warping constant of the gross cross-section;
$I_{w,sh}$	is	the value of I_w for a cross-section with sharp corners;
ϕ	is	the angle between two plane elements;
m	is	the number of plane elements;
n	is	the number of curved elements;
r_j	is	the internal radius of curved element j.

(5) The reductions given by expression (5.1) may also be applied in calculating the effective section properties A_{eff}, $I_{y,eff}$, $I_{z,eff}$ and $I_{w,eff}$, provided that the notional flat widths of the plane elements are measured to the points of intersection of their midlines.

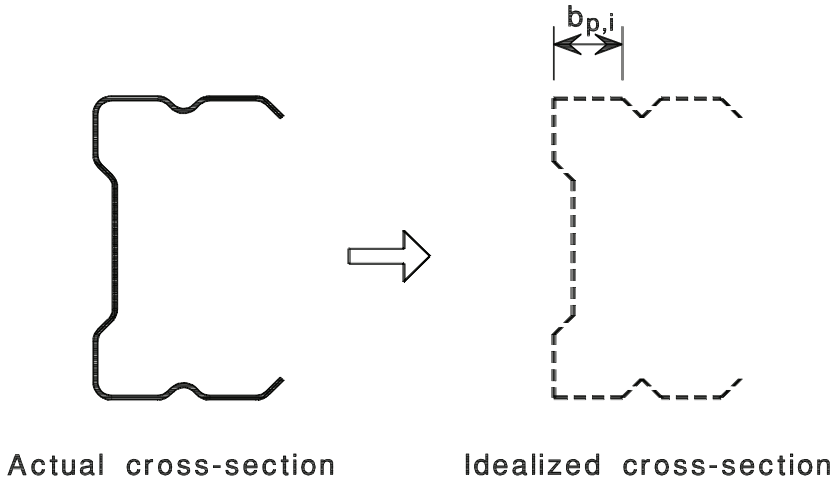


Figure 5.2: Approximate allowance for rounded corners

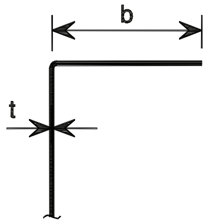
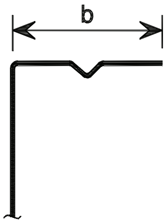
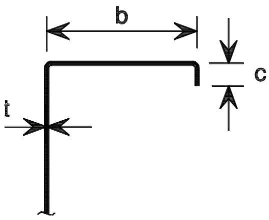
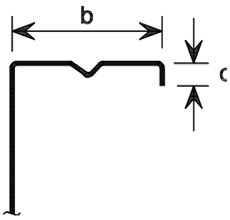
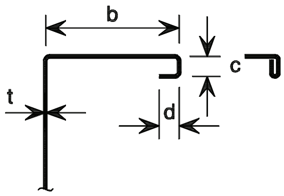
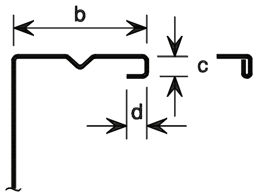
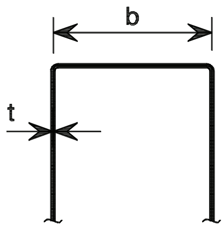
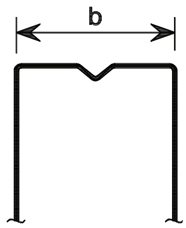
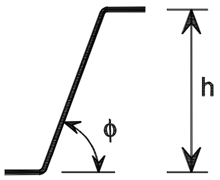
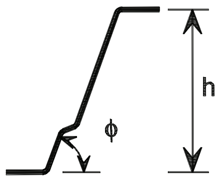
(6) Where the internal radius $r > 0,04\, t\, E / f_y$ then the resistance of the cross-section should be determined by tests.

5.2 Geometrical proportions

(1) The provisions for design by calculation given in this Part 1-3 of EN 1993 should not be applied to cross-sections outside the range of width-to-thickness ratios *b/t*, *h/t*, *c/t* and *d/t* given in Table 5.1.

NOTE: These limits *b/t*, *h/t*, *c/t* and *d/t* given in table 5.1 may be assumed to represent the field for which sufficient experience and verification by testing is already available. Cross-sections with larger width-to-thickness ratios may also be used, provided that their resistance at ultimate limit states and their behaviour at serviceability limit states are verified by testing and/or by calculations, where the results are confirmed by an appropriate number of tests.

Table 5.1: Maximum width-to-thickness ratios

Element of cross-section		Maximum value
		$b/t \leq 50$
		$b/t \leq 60$ $c/t \leq 50$
		$b/t \leq 90$ $c/t \leq 60$ $d/t \leq 50$
		$b/t \leq 500$
		$45° \leq \phi \leq 90°$ $h/t \leq 500 \sin\phi$

(2) In order to provide sufficient stiffness and to avoid primary buckling of the stiffener itself, the sizes of stiffeners should be within the following ranges:

$$0,2 \leq c/b \leq 0,6 \qquad \dots (5.2a)$$

$$0,1 \leq d/b \leq 0,3 \qquad \dots (5.2b)$$

in which the dimensions b, c and d are as indicated in table 5.1. If $c/b <$ 0,2 or $d/b <$ 0,1 the lip should be ignored ($c = 0$ or $d = 0$).

NOTE 1: Where effective cross-section properties are determined by testing and by calculations, these limits do not apply.

NOTE 2: The lip measure c is perpendicular to the flange if the lip is not perpendicular to the flange.

NOTE 3: For FE-methods see Annex C of EN 1993-1-5.

5.3 Structural modelling for analysis

(1) Unless more appropriate models are used according to EN 1993-1-5 the elements of a cross-section may be modelled for analysis as indicated in table 5.2.

(2) The mutual influence of multiple stiffeners should be taken into account.

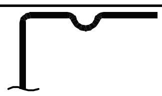
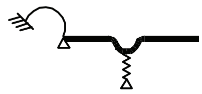
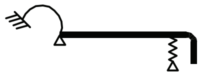
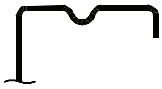
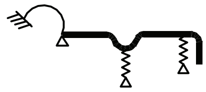
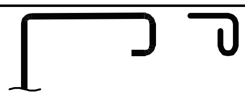
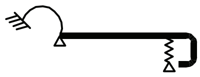
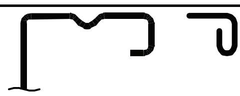
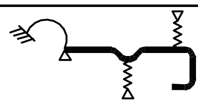
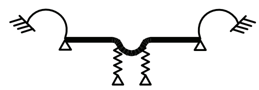
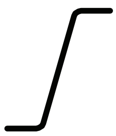
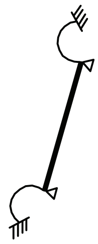
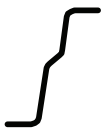
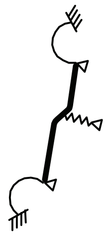
(3) Imperfections related to flexural buckling and torsional flexural buckling should be taken from table 5.1 of EN 1993-1-1

NOTE: See also clause 5.3.4 of EN 1993-1-1.

(4) For imperfections related to lateral torsional buckling an initial bow imperfections e_0 of the weak axis of the profile may be assumed without taking account at the same time an initial twist

NOTE: The magnitude of the imperfection may be taken from the National Annex. The values e_0/L = 1/600 for elastic analysis and e_0/L = 1/500 for plastic analysis are recommended for sections assigned to LTB buckling curve a taken from EN 1993-1-1, section 6.3.2.2.

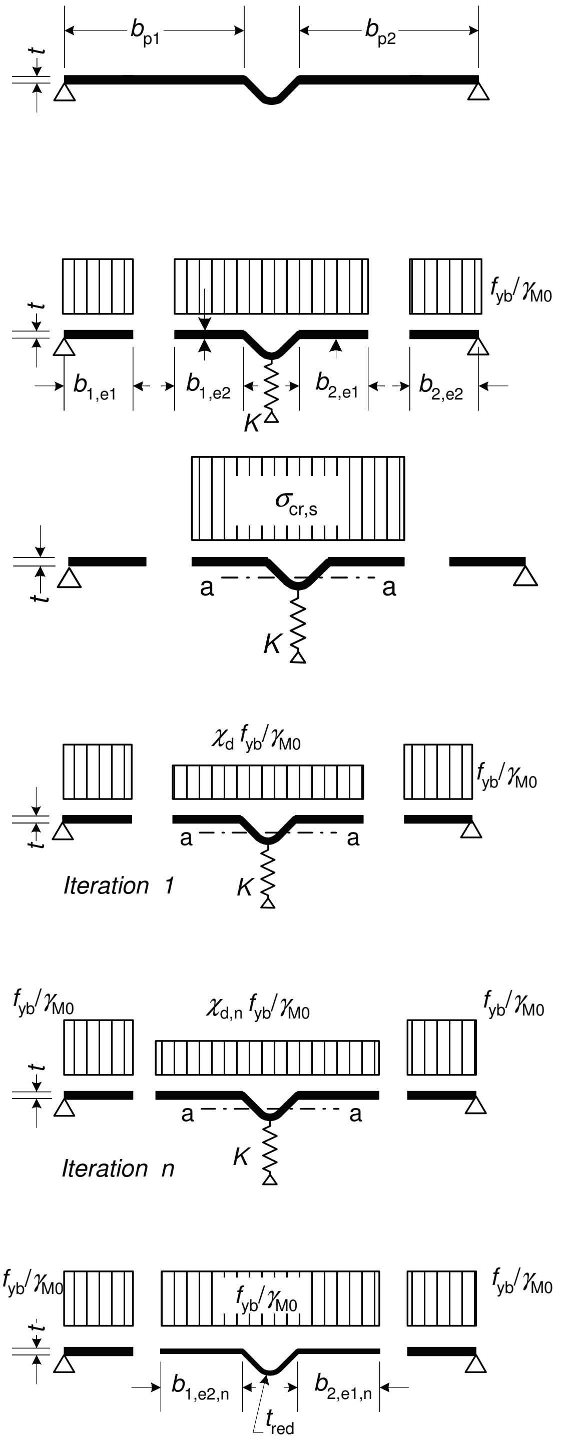
Table 5.2: Modelling of elements of a cross-section

Type of element	Model	Type of element	Model
			
			
			
			
			

5.4 Flange curling

(1) The effect on the loadbearing resistance of curling (i.e. inward curvature towards the neutral plane) of a very wide flange in a profile subjected to flexure, or of a flange in an arched profile subjected to flexure in which the concave side is in compression, should be taken into account unless such curling is less than 5% of the depth of the profile cross-section. If curling is larger, then the reduction in loadbearing resistance, for instance due to a decrease in the length of the lever arm for parts of the wide flanges, and to the possible effect of the bending of the webs should be taken into account.

NOTE: For liner trays this effect has been taken into account in 10.2.2.2.



a) Gross cross-section and boundary conditions

b) **Step 1**: Effective cross-section for $K = \infty$ based on $\sigma_{com,Ed} = f_{yb} / \gamma_{M0}$

c) **Step 2**: Elastic critical stress $\sigma_{cr,s}$ for effective area of stiffener A_s from step 1

d) Reduced strength $\chi_d f_{yb} / \gamma_{M0}$ for effective area of stiffener A_s, with reduction factor χ_d based on $\sigma_{cr,s}$

e) **Step 3**: Optionally repeat step 1 by calculating the effective width with a reduced compressive stress $\sigma_{com,Ed,i} = \chi_d f_{yb} / \gamma_{M0}$ with χ_d from previous iteration, continuing until $\chi_{d,n} \approx \chi_{d,(n-1)}$ but $\chi_{d,n} \leq \chi_{d,(n-1)}$.

f) Adopt an effective cross-section with $b_{1,e2}$, $b_{2,e1}$ and reduced thickness t_{red} corresponding to $\chi_{d,n}$

Figure 5.10: Compression resistance of a flange with an intermediate stiffener

(2) Calculation of the curling may be carried out as follows. The formulae apply to both compression and tensile flanges, both with and without stiffeners, but without closely spaced transversal stiffeners at flanges. For a profile which is straight prior to application of loading (see figure 5.3).

$$u = 2\frac{\sigma_{\mathrm{a}}^{2}}{E^{2}}\frac{b_{\mathrm{s}}^{4}}{t^{2}z} \qquad \text{... (5.3a)}$$

For an arched beam:

$$u = 2\frac{\sigma_{\mathrm{a}}}{E}\frac{b_{\mathrm{s}}^{4}}{t^{2}r} \qquad \text{... (5.3b)}$$

where:

- u is bending of the flange towards the neutral axis (curling), see figure 5.3;
- b_{s} is one half the distance between webs in box and hat sections, or the width of the portion of flange projecting from the web, see figure 5.3;
- t is flange thickness;
- z is distance of flange under consideration from neutral axis;
- r is radius of curvature of arched beam;
- σ_{a} is mean stress in the flanges calculated with gross area. If the stress has been calculated over the effective cross-section, the mean stress is obtained by multiplying the stress for the effective cross-section by the ratio of the effective flange area to the gross flange area.

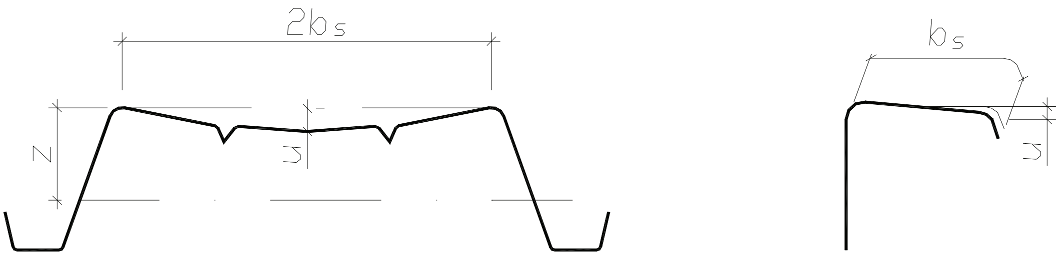


Figure 5.3: Flange curling

5.5 Local and distortional buckling

5.5.1 General

(1) The effects of local and distortional buckling should be taken into account in determining the resistance and stiffness of cold-formed members and sheeting.

(2) Local buckling effects may be accounted for by using effective cross-sectional properties, calculated on the basis of the effective widths, see EN 1993-1-5.

(3) In determining resistance to local buckling, the yield strength f_{y} should be taken as f_{yb} when calculating effective widths of compressed elements in EN 1993-1-5.

NOTE: For resistance see 6.1.3(1).

(4) For serviceability verifications, the effective width of a compression element should be based on the compressive stress $\sigma_{\mathrm{com,Ed,ser}}$ in the element under the serviceability limit state loading.

(5) The distortional buckling for elements with edge or intermediate stiffeners as indicated in figure 5.4(d) are considered in Section 5.5.3.